

Ch.12 Getting the Message

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Dirty Faces

Knowledge

Theorems and Possibility Sets

Back to Dirty Faces

Dirty Faces

- ▶ A , B and C travel together in a railway carriage and each one has a dirty face.
- ▶ If one is conscious of appearing in public with a dirty face, she will surely blush.
- ▶ Nobody blushes. $\Rightarrow$ None of them knows that her own face is dirty, although each can clearly see the dirty faces of the others.
- ▶ A clergyman who always tells the truth enters the carriage and announces that some one has a dirty face.
- ▶ A blushes and why?

- ▶ *A*: Suppose that my face were clean. Then *B* would reason as follows:
- ▶ *B*: I see that *A*'s face is clean. Suppose that my face were also clean. Then *C* would reason as follows:
- ▶ *C*: I see that *A*'s and *B*'s faces are clean. If my face were clean, nobody's face would be dirty. But the clergyman announced..., so my face is dirty, and I must blush.
- ▶ *B*: Since *C* hasn't blushed, my face is dirty. So I must blush.
- ▶ *A*: Since *B* hasn't blushed, my face is dirty. So I must blush.

Knowledge

- ▶ state of world, ω
- ▶ knowledge operator, K
- ▶ possibility operator, P
- ▶ axioms about K and P
- ▶ truism

State of World

- ▶ In the example of the dirty faces, there are 8 possible states of world.
- ▶ Let C denote a clean face and let D denote a dirty face. The 8 states of worlds are:

	1	2	3	4	5	6	7	8
A	C	D	C	C	D	D	C	D
B	C	C	D	C	D	C	D	D
C	C	C	C	D	C	D	D	D

- ▶ state $\omega = 1$: everyone has a clean face.
- ▶ universe $\Omega \equiv \{1, 2, 3, 4, 5, 6, 7, 8\}$
- ▶ event E is a subset of Ω
 - ▶ If the true state is an element of an event E , then E has occurred.
 - ▶ The event that A has a dirty face, $D_A = \{2, 5, 6, 8\}$.
 - ▶ If $\omega = 8$, then D_A has occurred.

Knowledge Operator, K

	1	2	3	4	5	6	7	8
A	C	D	C	C	D	D	C	D
B	C	C	D	C	D	C	D	D
C	C	C	C	D	C	D	D	D

- ▶ KE is the set of states of the world in which A knows that E has occurred.
- ▶ $D_A = \{2, 5, 6, 8\}$
- ▶ The event that A knows her own face being dirty, $KD_A = ?$
 - ▶ After the clergyman's announcement, and before A reaches any deep induction, she looks at two other faces and only when two other ladies both have clean faces, could A know she herself has a dirty face. $KD_A = \{2\}$.
 - ▶ But, $\omega = 8 \notin KD_A$, so KD_A does not occur and A does not know her face is dirty yet.

Possibility Operator, P

- ▶ PE means the event that A considers the event E possible.
- ▶ $PE \equiv \sim K \sim E$

Axioms

$$(K0) K\Omega = \Omega$$

$$(P0) P\emptyset = \emptyset$$

$$(K1) K(E \cap F) = KE \cap KF$$

$$(P1) P(E \cup F) = PE \cup PF$$

$$(K2) KE \subseteq E^1$$

$$(P2) E \subseteq PE$$

$$(K3) KE \subseteq K(KE) \equiv K^2E$$

$$(P3) P^2E \subseteq PE$$

$$(K4) PE \subseteq KPE$$

$$(P4) PKE \subseteq KE$$

- ▶ There appear to be 10 different axioms.
- ▶ But since K axioms $\Leftrightarrow P$ axioms, there are only 5 effective axioms.
- ▶ For instance, we can derive (P0) from (K0):

$$\begin{aligned} P\emptyset &\equiv \sim K \sim \emptyset \\ &= \sim K\Omega \\ &= \sim \Omega \quad (\text{from (K0)}) \\ &= \emptyset \quad \square \end{aligned}$$

¹ $P \subseteq Q$ means if P then Q .

Truism

- ▶ Def: The event T is a truism for A , if T can't be true w/o A knowing it, i.e. $T \subseteq KT$.
- ▶ Recall (K2) $KE \subseteq E$, so for a truism T , we have $T = KT$.

Theorems and Possibility Sets

- ▶ We shall establish 3 theorems that lead us to characterize when A knows her face is dirty.
- ▶ During the course, we need to define a new concept, possibility set.

Theorem 1: A knows E has occurred iff a truism that implies E has occurred.

Proof: First we'll establish $\Rightarrow$.

For this purpose, we have to show $\exists T, KE \subseteq T$ where T is a truism and $T \subseteq E$.

Define $T \equiv KE$.

Clearly, $KE \subseteq T$.

From (K3), $T \subseteq KT$, T is a truism. $((K3) KE \subseteq K(KE))$

From (K2), $T \subseteq E$. $((K2) KE \subseteq E)$

We'll now establish $\Leftarrow$.

Suppose T is a truism and $T \subseteq E$. We need to show that $T \subseteq KE$.

We'll first establish that if $G \subseteq H$, then $KG \subseteq KH$.

Recall from (K1), $K(G \cap H) = KG \cap KH$.

If $G \subseteq H$, we have $KG = KG \cap KH$ which implies $KG \subseteq KH$.

From this, $T \subseteq E \Rightarrow KT \subseteq KE$.

Recall that T is a truism, $T \subseteq KT \subseteq KE$. $\square$

Theorem 2: $P\{\omega\}$ is the smallest truism containing ω .

Lemma 1: If $E \subseteq F$, $PE \subseteq PF$.

Proof: From (P1), $P(E \cup F) = PE \cup PF$

If $E \subseteq F$, $P(E \cup F) = PF = PE \cup PF$, $PE \subseteq PF$. $\square$

Proof of Theorem 2: From (P2), $E \subseteq PE$, so $\{\omega\} \subseteq P\{\omega\}$, i.e. $P\{\omega\}$ contains ω .

From (K4), $PE \subseteq KPE$, i.e. $\forall E$, PE is a truism.

Lastly, consider any truism T containing ω , we'll show $P\{\omega\} \subseteq T$.

From lemma 1, $\{\omega\} \subseteq T \Rightarrow P\{\omega\} \subseteq PT = PKT \subseteq KT = T$.

The equalities are due to T being a truism; the $\subseteq$ is due to (P4). $\square$

Possibility Set

- ▶ The last theorem involves possibility sets.
- ▶ Def: The possibility set $P(\omega)$ is the set of states that A considers possible when the true state is ω .
- ▶ Distinction between $P\{\omega\}$ and $P(\omega)$:
 - ▶ $a \in P(\omega) \Leftrightarrow \{\omega\} \subseteq P\{a\}$
 - ▶ $a \in P\{\omega\} \Leftrightarrow \{a\} \subseteq P\{\omega\}$
- ▶ But, $P\{\omega\} = P(\omega)$!
- ▶ To prove this, from above, it's sufficient to show that
 $\forall$ states $c, d, \{c\} \subseteq P\{d\} \Rightarrow \{d\} \subseteq P\{c\}$

Lemma 2: $\{c\} \subseteq P\{d\} \Rightarrow \{d\} \subseteq P\{c\}$.

Proof: Suppose $\{c\} \subseteq P\{d\}$, but $\{d\} \not\subseteq P\{c\}$.

Then $\{d\} \subseteq \sim P\{c\} \equiv K \sim \{c\}$.

Applying lemma 1, $\{c\} \subseteq P\{d\} \subseteq PK \sim \{c\} \subseteq K \sim \{c\}^2 \equiv \sim P\{c\}$.

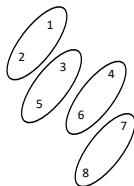
It contradicts (P2). $\square$

Possibility Set

- ▶ Def: The possibility set $P(\omega)$ is the set of states that A considers possible when the true state is ω .
- ▶ Recall that there are 8 states in the example of dirty faces.

	1	2	3	4	5	6	7	8
A	C	D	C	C	D	D	C	D
B	C	C	D	C	D	C	D	D
C	C	C	C	D	C	D	D	D

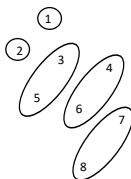
- ▶ For A , what are $P_A(1), \dots, P_A(8)$ before the clergyman appears?
- ▶ $P_A(1) = \{1, 2\} = P_A(2) \dots$ which are expressed in the following figure:



- ▶ Possibility sets change when new information arrives.
- ▶ The clergyman will announce that someone has a dirty face iff it is true.
- ▶ What are A 's possibility sets right after the clergyman arrives?

	1	2	3	4	5	6	7	8
A	C	D	C	C	D	D	C	D
B	C	C	D	C	D	C	D	D
C	C	C	C	D	C	D	D	D

- ▶ $P_A(1) = \{1\}$, $P_A(2) = \{2\}$ and all other possibility sets remain the same.



- ▶ According to this figure, does A know $D_A (= \{2, 5, 6, 8\})$ has occurred?

- ▶ Theorem 1: A knows E has occurred iff a truism that implies E has occurred.
- ▶ Theorem 2: $P\{\omega\}$ is the smallest truism containing ω .
- ▶ Theorem 3: A knows that E has occurred in state ω iff $P(\omega) \subset E$.

Proof: $\Rightarrow$

If A knows that E has occurred in state ω , from Theorem 1, $\exists$ truism T , $\omega \in T \subset E$.

From Theorem 2, $\omega \in P\{\omega\} = P(\omega) \subseteq T \subseteq E$.

⇐

From Theorem 2, in state ω , the truism $P(\omega)$ occurs.

If $P(\omega) \subset E$, from Theorem 1, A knows that E has occurred in state ω . $\square$

- Back to the previous slide, because $P_A(8) = \{7, 8\} \not\subseteq D_A = \{2, 5, 6, 8\}$, according to Theorem 3, A does not know that she has a dirty face right after the clergyman's announcement.

Possibility Set

- ▶ Def: C is a partition of Ω , if C is a collection of non-empty subsets of Ω such that every state in Ω is in exactly one of these subsets.
- ▶ ex. $C = \{\{1, 2\}, \{3, 4, 5\}, \{6, 7, 8\}\}$
- ▶ Lemma 3: $\forall \omega' \in P(\omega), P(\omega') = P(\omega)$

Proof: Suppose $\omega' \in P(\omega)$.

We'll first show that $\forall \omega'' \in P(\omega'), \omega'' \in P(\omega)$.

By the definition of a possibility set, $\omega'' \in P(\omega) \Leftrightarrow \{\omega\} \subseteq P\{\omega''\}$.

We have:

$$\{\omega\} \subseteq P\{\omega'\} \subseteq PP\{\omega''\} \subseteq P\{\omega''\}$$

The 1st $\subseteq$ follows from $\omega' \in P(\omega)$; the 2nd from $\omega'' \in P(\omega')$ which means $\{\omega'\} \subseteq P\{\omega''\}$, and lemma 1; the 3rd from (P3).

We still need to show that $\forall \omega'' \in P(\omega), \omega'' \in P(\omega')$, or to show that $\{\omega'\} \subseteq P\{\omega''\}$. This part is left as your exercise.

Possibility Set

Theorem 4: $\{P(\omega) | \omega \in \Omega\}$ is a partition of Ω .

Proof: From Theorem 2, $\omega \in P(\omega)$, hence $\bigcup_{\omega \in \Omega} P(\omega) = \Omega$.

Next, we'll show that $\forall \omega_1, \omega_2, P(\omega_1) \neq P(\omega_2), P(\omega_1) \cap P(\omega_2) = \emptyset$.

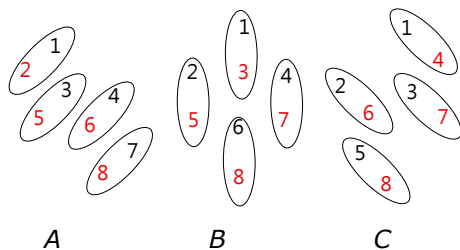
Suppose $P(\omega_1) \neq P(\omega_2)$ and $\exists \omega_3 \in P(\omega_1) \cap P(\omega_2)$.

From lemma 3, $P(\omega_1) = P(\omega_3) = P(\omega_2)$ which contradicts the presumption. $\square$

Def: Consider 2 partitions of Ω , C and D . C is a refinement of D if each set in C is a subset of a set in D .

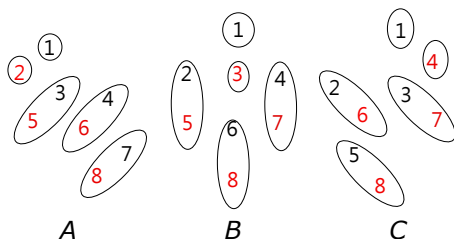
Ex. A 's information partition becomes refined after the clergyman's announcement.

Information Partitions before the Clergyman's Announcement



	1	2	3	4	5	6	7	8
A	C	D	C	C	D	D	C	D
B	C	C	D	C	D	C	D	D
C	C	C	C	D	C	D	D	D

After the Clergyman's Announcement



No one knows that her face is dirty because $P_i(8) \not\subseteq D_i, i = A, B, C$.

Taking Turns to Blush: $A \rightarrow B \rightarrow C \rightarrow A \dots$

